

Multiple Choice (4 marks)

1. A single-electron atom has the electron in the $l = 2$ state. The number of allowed values of the quantum number m_l is
 - (a) 5
 - (b) 4
 - (c) 3
 - (d) 2
2. A grating spectrometer can just barely resolve two wavelengths of 500 nm and 502 nm, respectively. The number of slits in the grating is
 - (a) 2
 - (b) 250
 - (c) 5,000
 - (d) 10,000

True / False (26 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'The emission spectrum of the doubly ionized lithium atom Li^{2+} is similar to that of hydrogen' is True.
2. True or False: The correct answer to 'An organ pipe, closed at one end and open at the other, is capable of supporting a standing wave of any frequency' is False.
3. True or False: The correct answer to 'If the total energy of a particle of mass m is equal to twice its rest energy, then its momentum is $\sqrt{3}mc$ ' is True.
4. True or False: The correct answer to 'Electromagnetic radiation emitted from a nucleus is most likely in the gamma-ray region' is True.
5. True or False: The correct answer to 'A particle decays in 2.0 ms in its rest frame. If the same particle is moving at $0.8c$ relative to an observer, the observer measures the lifetime to be 3.2 ms' is True.
6. True or False: The correct answer to 'For which of the following thermodynamic processes is the work done by the system equal to the heat added to the system?' is 'isobaric expansion'.
7. True or False: The correct answer to 'The negative muon, μ^- , has properties most similar to the electron' is True.
8. True or False: The correct answer to 'An atom has filled $n = 1$ and $n = 2$ levels. How many electrons can it hold?' is 10.
9. True or False: The correct answer to 'A three-dimensional harmonic oscillator is in thermal equilibrium with a reservoir at temperature T . The probability of finding the oscillator in the ground state is $e^{-3/(k_B T)}$ ' is True.
10. True or False: The correct answer to 'De Broglie hypothesized that the linear momentum and wavelength of a particle are related by $p = h/\lambda$ ' is True.
11. True or False: The correct answer to 'The eigenvalues of a Hermitian operator are always real' is True.
12. True or False: The correct answer to 'According to the BCS theory, the attraction between Cooper pairs is mediated by phonons' is True.
13. True or False: The correct answer to 'The speed of light inside of a nonmagnetic dielectric material is less than c ' is True.

End of Paper